

§3.2

Addition in Other Bases

The same two methods you already use, with the regrouping rule changed

Today's Goal

After completing this section, you should be able to:

- Add single-digit numbers in a base other than ten by building an addition table for that base
- Add multi-digit numbers in a base other than ten using the standard algorithm
- Use the Exploding Dots model to add numbers in a base other than ten

Learning Target 9

LT.9 Different Number Bases

I can convert numbers from base ten to base n and vice versa for any whole number $n > 1$. I can add, subtract, multiply, and divide in different number bases.

Build the Table

Fill in every sum as a base 5 numeral. Base 5 has only the digits 0, 1, 2, 3, 4.

+	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	10
2	2	3	4	10	11
3	3	4	10	11	12
4	4	10	11	12	13

Adding 0 changes nothing.

Sums below 5 look the same as in base ten.

Every pair that makes 5 is 10.

Every other sum is one five and what is left over.

Regrouping as You Go

Evaluate $342_{\text{five}} + 134_{\text{five}}$ using the standard algorithm.

Column	Add	Base 5	Write	Carry
ones	$2 + 4$	11_{five}	1	1
fives	$1 + 4 + 3$	13_{five}	3	1
25s	$1 + 3 + 1$	10_{five}	0	1
125s	1	1	1	—

The Same Sum on the Machine

 $1 \leftarrow 5$

342_{five}			
$+134_{\text{five}}$			
After adding			
1s explode			
5s explode			
25s explode			

 $= 1031_{\text{five}}$

Which Would You Rather Use?

Both gave the same sum.

So what is different?

The standard algorithm regroups *as you go*, so you need every single-digit base 5 sum in your head.

Exploding Dots adds first and regroups *at the end*, so you only count dots and share out groups of 5.

A Puzzle in Base Five

This addition is in base 5. Each letter is a different digit, 0 to 4.

$$\begin{array}{r} \\ \\ + \\ \hline \\ \\ \\ \end{array}$$

Which digit is each letter?

Five digits, four letters, so one digit goes unused. Start at the ones.

Solving the Puzzle

One set of digits works: $A = 1, B = 3, C = 2, D = 4$.

$$\begin{array}{r}
 A B B \\
 + C C D \\
 \hline
 D A C
 \end{array}$$

$$\begin{array}{r}
 1 1 \\
 1 3 3 \\
 + 2 2 4 \\
 \hline
 4 1 2
 \end{array}$$

So $133_{\text{five}} + 224_{\text{five}} = 412_{\text{five}}$, and 0 is never used.

One Question to Leave With

**In base 5, why can
a carry never be 2?**

Start at the ones. Nothing carries into it, so the most it can hold is $4 + 4 = 8$. That is one group of five with 3 left over, so it carries 1.

So the next column takes in 1. The most it can hold is $4 + 4 + 1 = 9$, which is one group of five with 4 left over, so it carries 1 as well.

And that repeats: each column passes on 1, so the next takes in 1. Never 2.